The Causes and Effects of Video Game Addiction

on Teens and Young Adults

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Introduction

Providing Context

No words can express the joy that an artist feels when their work brings happiness to their audience, but it is also the artist's duty to ensure that no harm comes from it. As a game developer, the discussion of video game addiction has always been fascinating to me. We as developers and programmers strive to make our games fun as well as enjoyable. Yet, when do we cross the line in making our games too fun to the point of addiction? Every artist wants their work to be enjoyed, but not to ruin lives.

During my time as a student in Humber College's Game Programming course, I learned many aspects of game development such as AI systems and 3D rendering, but game ethics is not a topic openly discussed. So, I was quite surprised to keep finding myself as the only game programming student in philosophy and ethics elective courses at Humber. Among my classmates, writing and philosophy were seen as taboo, with wisdom and programming having no correlation. Of course, I disagreed, with gaming addiction being my primary argument to why ethics in gaming is just as important as ensuring a piece of code works.

Prior to Humber College, I took a Game Developing course in Centennial College to get my footing into the world of game deployment. Unfortunately, I saw many of my fellow students forgoing assignments in favor of playing video games. I always found it a little ironic that students would rather be playing games than making them, but I attributed this to the school's somewhat relaxed environment. I myself sometimes experienced long hours of playing games especially at this time, but seeing as what it did to my colleagues, I never take it to the point of forgoing duties. Nevertheless, I feel my unique insight as a game developer with an open mind can help bring a new perspective into the topic of gaming addiction.

Furthermore, video game addiction is a relatively new disorder, with Gaming disorder officially being recognized in 2018. We still have so much to learn between the relationship of gamers and addictive behaviors. Now in 2025, the notion of incorporating ethics in video game development to address video game addiction is much more prevalent. Video game addiction in particular is problematic in places like China and is arguably growing in the United States but also in Canada. Canada is currently seeing a fast-growing game industry, but this has the negative effect of more games being outputted and thus more people playing games in general. As more rehab centers start including treatment for gaming disorder in Canada as a result of gaming addiction, studying this topic is now more important than it has been since my time in Humber and Centennial.

Research Question

Therefore, I ask, what aspects of a video game make them hard to stop playing and addictive to a player? My research topic seeks to describe as well as explore what specific elements of video games contribute to video game addiction for young adults and teens. When does pleasure become pain? When does joy turn to poison? When does the game developer stop being an artist, and instead turns into a serpent?

Project Overview

Literature Review

This section will cover all pre-existing studies used in this paper thoroughly.

Addictive and Non-Addictive Game Mechanics

Covering studies that highlight video game addiction by how they are driven through the use of game mechanics.

Gaming & Gambling Addiction

Explores the links between gaming and gambling addiction as well as its harmful effects.

Ethics of Game Design

Will bring forth articles studies that question the morality of using addictive game mechanics within video games.

Methodology

Next, this section will go over how the research data in the 'Literature Review' is being used to answer my research question, how as well as where I found the data, and why I chose it.

Findings and Discussion

Finally, the overall findings of the studies and how they answer the research question will be discussed. This section also serves as notes for the accompanying presentation.

Literature Review

Research Question:

What aspects of a video game make them hard to stop playing and addictive to a player? My research topic seeks to describe as well as explore what specific elements of video games contribute to video game addiction for young adults and teens. In this research paper, I will examine the correlation between game addiction and other forms of addiction, primarily gambling, and how video games are engineered to keep players hooked as well as what game mechanics are the most addictive to a gamer.

In addition, this paper also seeks to define an ethical methodology for gaming addiction in a way that the general public can better understand it as well as address some of its misconceptions. Overall, this paper will present the effects of gaming addiction on teens (ages 13

to 17) and young adults (ages 18 to 24) as well as identify what aspects of a video game make them hard to stop playing to an addicted player.

Subheading One: Addictive and Non-Addictive Game Mechanics

When examining any addiction, it is important to consider how a person is affected individually to better understand the addiction's dangerous and put forth solutions. In terms of gaming, analyzing what game mechanics are addictive and non-addictive can lead us to better understand how they hook players in. With this in mind, addictive game mechanics are especially harmful for teens and young adults due to the negative effects it can have socially and mentally on a person. Unrestricted gaming access has been linked to increased anger and aggression as well as other negative traits. In addition, an individual can suffer from a lack of self-control when addicted to games, made worse by social media where it can become harder to get away from a virtual social environment. A game though its mechanics has the harmful power to isolate a person to the point where said game becomes their only recognizable social space. This will only make the addict want to play more as well as become angrier, depressed, and socially disconnected. Moreover, game mechanics that focus on removing the player from reality are a driving force for causing game addiction.

According to Yilmaz and Tunca, there is a direct correlation between time perception and gaming addiction. They "found that the flow experience experienced by online gamers and the increase in the level of distortion in time perception had high effects on game addiction" (Yilmaz & Tunca, 2024, para. 1). By distorting a player's sense of time, the user gets hooked into the game's virtual world and loses track of reality. Yilmaz and Tunca (2024) go on to say, "preventing individuals' time perception from deteriorating or taking preventive measures in terms of flow experience will prevent game addiction" (para. 1). Thus, treatment of game

addiction should focus on breaking or resisting the user's immersion in the world. Overall, distortions that take the player away from reality can be a driving force for game addiction. This is an example of how certain game elements are used to keep players playing.

However, it is not just game mechanics that can lead to gaming addiction. According to Frida André and company, there is a social harm that unrestricted gaming can have as well as how social media and the internet do play a part in fostering addictive gaming disorders. To be more specific, they explain that spending more than four hours a day engaging online socially can increase the probability of gaming addiction by 5.3 times (André et al, 2020, p. 6). In addition, they also confirmation that gaming addiction has been associated with negative factors such as anxiety, depression, and loneliness as well as play into social media dependency. As they say, "The addictive gaming has been associated to anxiety, depression, social phobia/anxiety, low social competence, low self-esteem, sleeping disorders and loneliness. It has also been associated to other computer based behavioural addictions such as Internet addiction and Facebook addiction" (André et al, 2020, p. 2). This is direct evidence of how social media is contributing to gaming addiction and said addiction leading to negative factors like depression.

To conclude, gaming addiction can develop not just form game mechanics but from outside gaming sources like online social media as well. Just spending four hours of non-stop online social media is enough to increase the likelihood of developing game addiction. Regardless of how teens develop gaming addiction, once they are hooked, the game mechanics will try to keep them hooked. As well as try to take them out of our world. How gaming affects a person individually is crucial for understanding its addictive properties and addressing them.

Subheading Two: Gaming & Gambling Addiction

As video games become more developed, they have the potential to introduce aspects that are on par with gambling and its addictive qualities. Just like with casino games, video games try to stimulate a constant since of joy though the thrill of winning or at least being close to it. However, this thrill of winning can tie into the flow experience to foster addictive spending habits akin to that of gambling. For instance, any game that implements a reward-based system can incentivize the player to make real money purchases to obtain in game items. This is important to consider as the links between gaming and gambling can help to better understand game addiction though an already established lens.

To begin, an element of video games that is near identical to gambling is the reward-based system known as 'loot boxes'. According to the article "Does the loot box open the door to addiction? A case report of gaming disorder with high charges for loot box purchases", "The chance-based nature of loot boxes is similar to gambling. Obtaining the desired items randomly from loot boxes reinforces the reward system, leading to an addiction to purchasing them. Therapy should address the patient's motivation for purchasing loot boxes while considering individual developmental characteristics and guide the patient toward finding satisfaction and a sense of achievement in activities besides gaming" (Inaguma et al, 2024, para. 24). To begin with, the system of loot boxes is similar to gambling due to being a reward-based system that relies on random chance. Next, due to its random nature, desired items are not guaranteed which incentives the player to keep spending money on them thus forming an addiction. Consequently, treatment of this addiction should focus on adjusting the brain's reward system to find pleasure in other activities that do not involve gaming or making in game purchases. Game mechanics

like loot boxes that enforce gambling habits nevertheless provide the same dopamine rush as casino games such as slot machines.

In regards to treating addicted players, we can use vitality structures to better assess the primary issues of gaming addiction. Karhulahti (2024) states that there are "three common vitality structures (CLIMB, FINAL STRETCH, ALERT) that are relevant to videogames that treatment-seeking players have reported as sources of their problems" (para. 103). To start, CLIMB focuses on the player's constant progress towards a single goal. FINAL STRETCH focuses on a drive to keep going when almost reaching said goal. Lastly, ALERT focusses on the constant information feedback of notifying the player on the game's events. These vitality structures can be used as a starting point to help better understand the relationship between gamers and game designers. As Karhulahti (2024) states, "each vitality structure was shown to serve distinct functions in relation to differently orchestrated videogames (and non-videogames), thus allowing for specific construct-driven hypotheses on the nature of such designs and their players" (para. 103). With this in mind, each vitality structure is carefully implemented by the game designer for a distinct purpose.

To conclude, Inaguma brings to light many correlations between gambling and gaming, to the point where they are inseparable. In addition, Karhulahti's perspective on time and vitality structures is crucial for better understanding these addictive qualities at its core. Both casinos and video games try to provide the player with a constant stream of pleasure and rewards to encourage continued play while also distorting their sense of time. Games today have also normalized real money purchases which can developed into a bad spending habit if left unchecked. With loot boxes being the most direct example, as they utilize the same random chance mechanics of a slot machine. Like gambling, gaming can lead to financial troubles if left

unchecked, as well as effect a person both mentally and socially though their game mechanics. These same game mechanics can also be better explored using vitality structures that can be tied into the CLIMB, FINAL STRETCH or ALERT structures.

Subheading Three: Ethics of Game Design

Gaming addiction presents many hardships as well as challenges to teens and young adults as they are the most vulnerable demographic to gaming addiction. Yet despite this, some game developers choose to initially make their games as addictive as possible. To better assess what elements of a video game contribute to game addiction for young adults and teens, we must first explore how these addictive qualities take root, but more importantly, their ethics. It is easy to assume that adolescents become addicted to video games primarily out of blissful joy. While addiction to video games is also rooted in pleasure, another factor is the implementation of gameplay elements that seek to keep the player invested. This is highly unethical, games should be fun to play, but not to the point of them being almost like a digital drug. Moreover, if a game forces the player to keep playing even when unhappy, then it longer becomes a game but digital opium.

To begin, gaming addiction is not strictly rooted in pleasure as many would like to believe. While it is true that many people who suffer from game addiction are driven by pleasure, if the addiction is strong enough than they will keep playing regardless. According to Jun-Jie Hew and company, video game addiction does not always equate to happiness, even being able to suppresses it. In fact, even when faced with displeasure, gaming addiction will still keep gamers playing even when their happiness is suppressed. Hew "noticed that the state of flow might not always lead to happiness, especially when the gamers have been experiencing the state of flow for a relatively long term, resulting in negative moods such as frustration and

exhaustion eventually. This suggests that excessive video game playing could have intervened the effect of flow on subjective happiness of gamers" (Hew et al., 2024, p. 5). The longer the player plays, the more they would want to keep playing, even when unhappy. When the addiction is strong enough, gamers will keep playing even when unhappy, almost like a drug. To this end, if a game no longer becomes a game but a drug, can well really call it ethical game design?

In regards to ethics, not every game developer would agree with my research. I encourage all game developers to design their games ethically, but the truth is that some developers will prioritize profits over morals. In Christian Klemm and Wolter Pieters' ethical perspective, we see the case of the designers of a Massively Multiplayer Online Role-Playing game creating game mechanics that are ethically questionable and intentionally trying to get their players hooked. According to the authors, "MMORPG's core mechanics revolve around making the player play over a long period of time and keeping him or her busy so that the player won't consider other games or forms of entertainment. This intention is ethically questionable" (Klemm & Pieters, 2017, p. 88). Yet despite this, the authors also show an alternative approach to designing game mechanics by encouraging less addictive play time and thus much more ethical. One of the ways they suggested to make games less addicting is though automated processes. As the say, "To reduce the need for players to spend a lot of time on simple mindless tasks… the game could offer to perform one of the tasks while the player is actually offline… This could help minimize the player's feeling of falling" (Klemm & Pieters, 2017, p. 90). Overall, Klemm and Pieters show how some game designers choose profit over morals as well as how there are solutions to make game mechanics less addictive and more ethical overall. Thus, game developers are capable of making fewer addictive games, and we should stride to encourage it.

Ultimately, if gaming addiction is left unchecked, it can force players to keep playing even when unhappy, which will only make the addiction worse. While multiplayer games can bring people together, it can also tear people apart socially, as they do not want to leave the digital world. Unrestricted gaming access is a core component to gaming addiction, with those effected being unable to stop playing. Yet why should games be allowed to be designed in such a way? We are capable of creating game mechanics that elevate the stress of missing out or failing. Games should not be deigned like drugs, and to do so is highly unethical.

Process Reflection Questions

Question 1:

What is your most up-to-date version of your research question, incorporating any revision suggested by Prof. Tremblay?

My most up-to-date research question is "What aspects of a video game make them hard to stop playing and addictive to a player? My research topic seeks to describe as well as explore what specific elements of video games contribute to video game addiction for young adults and teens". This is almost unchanged from my previous research question as my core research has not changed. I did not receive any major revisions suggested by Prof. Tremblay.

Question 2:

What were the key changes you needed to make between the first draft of your literature review and the revised draft here? Why?

To be honest, my first draft really did not need any big changes, but rather a change of vision. I changed most of the headings and articles used as I felt they were redundant or not as relevant. I felt "Explaining Video Game Addiction" and "Effects of Gaming Addiction" was not needed because I would already be doing that passively. In addition, I really wanted go into the discussion of ethics and game mechanics within my research.

Question 3:

How did you decide upon the subheadings you chose when grouping your literature review articles together?

I knew off that bat I wanted to disuses the ethics of addictive game design as well as linking gaming addiction to gambling. Before I get into those subheadings, I needed an opening heading as well. Thus, I felt explaining the harm of video game mechanics and game addiction as a whole would be a good opener. Prior to making my headings, I already had my articles that discussed what I wanted to deeply talk about, and most of them thankfully fit into my vision.

Question 4:

What additional information would you have liked to include in your literature review to help your reader understand the context for your research, but do not have the time or resources to include in this literature review?

I personally would have loved to include human survey data collected by myself. I would include how long gamers play for, what they currently play, and how it has affected them. However, collecting such data would have proved to be very costly for both my money and time. I felt going out in to the real world to gather data from actual gamers would have been extremely beneficial to my research, but ultimately unfeasible. In addition, the Seneca REB would not have allowed human participants in my research, which I will respect.

Methodology

My research topic seeks to describe as well as explore what specific elements of video games contribute to video game addiction for young adults and teens. Thus, my research question is as follows, what aspects of a video game make them hard to stop playing and addictive to a player? To answer this question, I turned to academic journal articles as well as research studies for my data. In particular, I sought data that explored video game mechanics, length of time data, age, and the effects of too much screen time to a person. I also sought core links between gambling and gaming addiction to see if there is a coloration between these two.

To elaborate, I chose to focus on game mechanics, time, and gambling data as these have been best shown to best access addiction in gaming studies. Traditionally, many early gaming studies that tried to asses gaming addiction were plagued by a myriad of issues. Such issues

include the studies being biased by as well as paint gamers in a negative light. These research studies, particularly those carried in the wake of national school tragedies, were often carried out by intentionally attempting to invoke a hostile response from the participants. Such studies were also financed by pro-firearm organizations that sought to shift blame on gun violence to violent video games. Other issues include failing to find corrections with gaming addiction and other forms of addiction, as well as only viewing gaming dependency as strictly a response to pleasure. There were also issues where the researcher lacked sufficient knowledge on video games to carry out such studies.

To address these issues, many gaming studies that are highly creditable do not simply look at video games, but their mechanics as well. Allowing the researcher to better understand the game's inner workings, and not just their surface value. Highly creditable researchers that carried either have extensively played video games as a hobby, or even better, are game developers themselves. This greatly helps to alleviate bias and provides firsthand knowledge to these kinds of studies. Recently, there has also has been a push to view gaming addiction, though the lens of gambling addiction. This has proven to be very beneficial as gaming addiction is relatively new, but gambling has an extensive amount of existing research. Age must also be considered as teens and young adults are often more open to additive behaviors as oppose to adults and seniors.

Unfortunately, I came across many studies that fell short of my expectations. As mentioned before, researchers who lack a good amount of knowledge on video games often conduit poorer or biased studies on gaming addiction. I have come across a few of these studies that indicated a lack of video game knowledge, resulting in surface level research. Journal articles and research studies that were also outside my scope included articles that were far too

old, and articles that discussed board games as oppose to video games. Older gaming research studies have the risk of suffering from irrelevant data, as video games change, so too their additive properties. A game like Super Mario Bros will not have the same addictive qualities as a game like Call of Duty. Video games often progress not just technologically, but how they are designed as well. Too much screen usage has also been proven to affect the brain and behaviors in a person, not just from video games but television as well.

Furthermore, to find the Journal articles and research studies used in my project, I turned to Seneca Library for the most creditable sources that are peer-reviewed. I started by assembling a list of keywords that can help find the data I was looking for. These keywords include “gaming addiction”, “gaming and gambling”, “gaming screen time”, and “game mechanics addiction”. I researched these keywords on Seneca Library, reviewing the top ten article results by first reading their abstract and only giving further if they presented interesting claims or unique data to my research question. I knew studies that are relevant to my project would specifically focus on game mechanics, game usage, or be backed up with real world examples. If a study discussed addiction by how a game worked inside to get you hooked, it is a must have. The most invaluable keywords that greatly helped isolate my relevant studies proved to simply be “gaming addiction” and “game mechanics addiction”. I did have to reject a few studies due to inefficient data, bias, or being overall irrelevant. Some such studies I found and rejected only focused on older video games like various Wii games, or disused board games alongside video games which can risk skewing my project scope.

Findings and Discussion

Slide 1:

Title

- This is the start of the presentation
- What specific elements of a video game contribute to video game addiction for young adults and teens?
- How are games engineered to keep players playing?

Slide 2:

Research Question

What aspects of a video game make them hard to stop playing and addictive to a player?

- How are games engineered to keep players playing?
- What is the correlation between gaming addiction and other forms of addiction?
- How do we more ethically design games for young adults and teens?
- What are the effects of gaming addiction on teens and young adults?

Slide 3:

Addictive (& Non-Addictive) Game Mechanics - Findings

Introduction to this theme:

- Explain how game mechanics are designed to keep you hooked.
- It is important because game mechanics are the driving force for getting people hooked.
- Highlight the negative factors:
    - Anxiety
    - Depression
    - Loneliness
- Game mechanics are designed to keep you hooked and addicted.
- Gaming addiction has been associated with negative factors such as anxiety, depression, and loneliness which can play into other addictions as well.
- Time distortions are being used to fuel gaming addiction to force players into the “I'm winning I can't stop” mentality to get them hooked.

Quotes/paraphrasing:

Quote 1:

“The one factor showing the most powerful correlation with problematic or addictive gaming was, interestingly, spending more than four hours daily chatting on internet/social media,

increasing the probability 5.3 times. This subgroup also showed a disproportionate prevalence of both engaged, problem, and addicted gamers" (André et al, 2020, p. 6).

- André claims that spending more than four hours a day engaging online socially can increase the probability of gaming addiction by 5.3 times.
- This means anyone can become addicted to video games, even when not playing them!

Quote 2:

"It was found that the flow experience experienced by online gamers and the increase in the level of distortion in time perception had high effects on game addiction. In this context, it is suggested that preventing individuals' time perception from deteriorating or taking preventive measures in terms of flow experience will prevent game addiction" (Yilmaz & Tunca, 2024, p. 1).

- Yilmaz claims that the achievement-based nature of video game reward systems plays a part in fostering additive behaviors.

Quote 3:

"A positive and significant result was found between the self-efficacy and the achievement. It is possible to say that if the players feel competent, they affect the in-game achievement. Confident players achieve success in games. A positive and significant relationship was found between the curiosity and the flow experience. The state of wanting to see the next stage that takes place during the game affects the flow experience" (Yilmaz & Tunca, 2024, p. 9).

- This shows how desire can overcome our will to back out though conditioning and chasing rewards.
- We become slaves to desires.

Slide 4:

Addictive (& Non-Addictive) Game Mechanics - Statistics

Gaming addiction:

- 75% percent US households have at least one person who plays video games
- 70% of those in the household under 18 regularly play video games
- 3.05% of people in the world experience pathological gaming (3 in 100 people)
- Four hours of social media has a 5.3x chance to develop gaming addiction

Structural model proposal:

- 65% of adults participate in video games in the US alone
- 48.3% of the 331 participants were under 23 years of age for the survey
- 39.3% play for more than two hours a day without stopping
- Distortion in Time Perception had the second highest arithmetic score at 7.20

Slide 5

Addictive (& Non-Addictive) Game Mechanics – Discussion

Discussion of theme:

- What are the effects of gaming addiction on teens and young adults?
- It causes depression though social isolation and distortion of time perception
- It causes Anti-Social behavior, with the risk of it occurring up to 5.3 times

Summary conclusion:

- Theme highlights the social harm that unrestricted gaming can have as well as how social media and the internet play a part in fostering addictive gaming disorders.
- Theme also highlights the connection between time perception and gaming addiction as well as how adjusting the player's flow experience can help to prevent it.

Slide 6

Gaming & Gambling Addiction - Findings

Introduction to this theme:

- Links between gaming addiction and gambling are very strong.
- Data shows that designers use conditioning and reward systems to get players hooked by reinforcing the reward system.
- It is important because linking gaming to gambling will help us better understand it.
- Because it is relatively new, do not really understand gaming addiction, but we do understand gambling.
- Game mechanics like the 'loot box' are similar to gambling due to being a reward-based system that relies on random chance akin to casino games.
- Using vitality structures are needed to better assess the primary issues of gaming addiction.
- The fact that gaming and gambling are very similar is very scary.
- In a way, gaming should be treated like gambling.

Quotes/paraphrasing:

Quote 1:

"The designers of gambling games are known to have identified several mechanisms with a high probability of contributing to people's gambling behaviours. It was B.F. Skinner himself who suggested no later than in the 1950s that conditioning had already been efficiently integrated to gambling design" (Karhulahti, 2024, p. 4).

- Karhulahti claims that the Vitality structure framework can be used to better understand addictive qualities.

- Gaming should be investigated in more studies using these vitality structures.
- It is all about rewiring the brain and conditioning.

Quote 2:

"The chance-based nature of loot boxes is similar to gambling. Obtaining the desired items randomly from loot boxes reinforces the reward system, leading to an addiction to purchasing them. Therapy should address the patient's motivation for purchasing loot boxes while considering individual developmental characteristics and guide the patient toward finding satisfaction and a sense of achievement in activities besides gaming" (Inaguma et al, 2024, p. 5).

- Inaguma claims that video games have the potential to introduce aspects that are on par with gambling which can foster addictive spending habits.
- Gambling games are almost identical to video games, while they differ many ways, how they hook players in to keep them playing with rewards is the same.
- We should help patients by giving them other things to do.

Quote 3:

"His gaming behavior was likely to be the element of gambling inherent in the loot box. There is a strong resemblance between getting rare items in loot boxes and taking a gamble. As players randomly obtain items from loot boxes, they become compelled to purchase more loot boxes in the expectation that they will eventually obtain a desired item. This is known as the variable ratio reinforcement schedule. This powerful behavioral reinforcement mechanism underlies the addiction to gambling." (Inaguma et al, 2024, p. 3).

- It's a coinflip.
- Getting a rare item is like hitting the jackpot.
- Not getting rare items causes players to spend more.
- This is targeting their desires.

Quote 4:

"The addictive gaming has been associated to anxiety, depression, social phobia/anxiety, low social competence, low self-esteem, sleeping disorders and loneliness. It has also been associated to other computer based behavioural addictions such as Internet addiction and Facebook addiction" (André et al, 2020, p. 2).

Slide 7:

Gaming & Gambling Addiction - Notes

Vitality structures:

- Goals in video games are any rewards that a player works towards, such as reaching the highest score, getting the highest rank, or getting a certain item

- **CLIMB:** Constantly moving up to reach the goal, like ladders in ranked games
- **STRETCH:** The feeling of almost reaching the goal
- **ALERT:** Constant feedback of information to keep players focused on the goal
- These vitality structures are based off treatment-seeking players

Loot box case:

- The case follows an unemployed man in his early 20s
- The man spent several thousand dollars a month on a Gacha mobile game
- The man experienced 'withdrawal' in the form of wanting more playtime, he was only given one hour of playtime per day
- Treatment focused on distractions and rewarding restraint, such as gaining more overall playtime for resisting the urge to play

Slide 8

Gaming & Gambling Addiction - Discussion

Discussion of theme:

- What is the correlation between gaming addiction and other forms of addiction?
- Like gambling, constantly spending money on loot boxes can lead you to ruin
- There are many aspects of gambling that are akin to gaming, like the random chance rewards-based system.

Summary conclusion:

- Theme highlights the importance of the concept of vitality structures (climbing to gain the next reward) in gaming and how designers of gambling-based games perfected the art of targeting a player's gambling behaviors though conditioning.
- Theme also confirms how a loot box, a specific game mechanic, is very akin to gambling with it providing the same dopamine rush as well as reward system comparable to that of casino games like slot machines.

Slide 9

Game Design Ethics - Findings

Introduction to this theme:

- Forcing players to keep playing while unhappy, like a mental illness.
- This is important because not all people who make games are sun-shin and rainbows.
- Not every game developer would agree with my research, as some seek profit over morals.
- These developers design systems to keep players engaged even when their mental health is negatively affected by gaming.
- There are solutions to make game mechanics less additive and more ethical overall.

- Money Money Money, that is all they care about! Such is the way of capitalism.

Quotes/paraphrasing:

Quote 1:

"Besides, from the perspective of video games, Lafan et al. (2016) noticed that the state of flow might not always lead to happiness, especially when the gamers have been experiencing the state of flow for a relatively long term, resulting in negative moods such as frustration and exhaustion eventually. This suggests that excessive video game playing could have intervened the effect of flow on subjective happiness of gamers" (Hew et al., 2024, p. 5)

- Hew's source claims that gaming addiction will keep gamers playing even when their happiness is suppressed.
- Gaming addiction is not about happiness, it never was, it's about mental health.

Quote 2:

"Subsequently, the higher the flow experienced by mobile MOBA gamers, the likely they will develop online game addiction and yield subjective happiness. However, their subjective happiness would be suppressed when mobile MOBA gamers realize that they have spent excessive time playing, supporting the suppressor role of online game addiction" (Hew et al., 2024, p. 9)

- Long term play can result in burn out.
- Realizing this can make you unhappy.
- Like food and drugs, games can be used as a suppressant that takes away joy.

Quote 3:

"MMORPG's core mechanics revolve around making the player play over a long period of time and keeping him or her busy so that the player won't consider other games or forms of entertainment. This intention is ethically questionable" (Klemm & Pieters, 2017, p. 88).

- Klemm's source stresses how designers of a MMOS created game mechanics to intentionally get players hooked.
- They know what they are doing and they don't care.
- Games that force people to keep playing are not games, or ethnical.

Quote 4:

"To reduce the need for players to spend a lot of time on simple mindless tasks… the game could offer to perform one of the tasks while the player is actually offline… This could help minimize the player's feeling of falling behind when not being online and add a possibly interesting layer of management to the game. This would make the mechanics less coercive, and leave more choice to the player." (Klemm & Pieters, 2017, p. 90).

- Giving players more choice though catch-up mechanics can help ease addiction.

- Fear of missing out (FOMO) deliberately target people's fear of missing the game's content.
- Would could make games less Fear of missing out (FOMO) based by providing catch up mechanics.
- As if a gaming company would actually do that. $$$$$$$$$$$$$$$$$

Slide 10:

Game Design Ethics - Notes & Statistics

Suppressor Role of Online Game Addiction:

- **The Flow Theory:** A psychological state experienced by individuals where they are very involved in certain activities
- Sense of time is neglected when in the flow state
- Mobile Legends was used as the studied game, which has made $502.5 million
- Rational Addiction Theory: Addicts get less pleasure from continuous consumption overtime

Ethical perspective on the effects of MMORPGs:

- MMORPG players spend an average of 22.72 hours a week in game
- 70% of players had played for 10 hours straight at least once
- Some players (8%) even spend 40+ hours a week playing
- It is more ethical to design games with constraints in mind

Slide 11

Game Design Ethics - Discussion

Discussion of theme:

- How do we best explain Video Game Addiction for young adults and teens?
- Focus on healing by encouraging treatment akin to dealing with gambling addiction.
- Focus on understanding gaming addiction though ethical game design.
- Playing games does not always mean happiness.
- It is not ethical to keep players playing when unhappy.
- Why are we designing games like drugs?

Summary conclusion:

- Theme highlights how video game addiction does not always equate to happiness and even suppresses it. In fact, even when faced with displeasure, gaming addiction will still keep gamers playing even when their happiness is suppressed.
- Theme also shows how there is an alternative approach to designing game mechanics by encouraging less additive play time, making them much more ethical.

- Ethical game example: Satisfactory notifies players to take a break when they play for more than 2 hours
- Ethical game example: Marvel Rivals uses battle passes that do not expire upon purchase, many other games instead have battle passes that expire in a certain which reinforces the fear of missing out (FOMO)
- Battle passes are a reward track in video games that reward items upon reaching certain goals, with a premium track that requires real money to unlock

Conclusion

Summary of Findings

Overall, it was found that gaming addiction is not strictly rooted in pleasure, ethical, or even isolated. There are strong links between gaming addiction and gambling as well as proof that certain game mechanics purposely designed to keep players hooked and addicted. Such mechanics work by invoking the winning mentality, distorting time, and preying on a player's need for rewards, isolating them socially. Yet, it is not ethical to keep players playing even when unhappy, these are not how games should be designed.

Furthermore, gaming addiction on young adults and teens has been associated with negative factors such as anxiety, depression, and loneliness which can play into other addictions as well. Time distortions are also being used to fuel gaming addiction to force players into the "I'm winning I can't stop" mentality in order to get them hooked. In regards to gambling, video games have the potential to introduce aspects that are on par with gambling which can foster addictive spending habits. To be understand these addictions, the vitality structure framework can be used to better assess these addictive qualities. Thus, it is important to keep in mind that not every game developer would agree with this kind of research, as some seek profit over morals. These developers design systems to keep players engaged even when unhappy.

However, there are solutions to make game mechanics less addictive and more ethical overall, such as incorporating checkup mechanics.

Acknowledgment of Limitation

Unfortunately, no study is perfect, and this this paper is no exception. Topics that assess gaming addiction by looking into game mechanics are few and limited. There are not as many researchers who are as proficient as game developers in terms of technical knowledge. I was also surprised to find that there are not many topics discussing the similarities between gambling and gaming, despite having very strong links. I suspect this likely due this kind of study requires knowledge in both the area of gambling and gaming. Despite the studies I found being very good, I wished there were more studies out there that discussed game mechanics and gambling to further strengthen my research.

Future Research

Finally, I want to stress that I have unfortunately seen many people, some of them being friends, succumb to video game addiction to the point of forgoing work. As mentioned previously I have seen game developers themselves choosing to play video games as oppose to making them. In my eyes, this is a possible epidemic, and more studies must be made into this topic.

Moreover, game addiction gaming addiction is not going away, it might even be in the same norm as drug addiction and alcoholism. Gaming addiction has been studied even before being classified back in 2018, but is primarily studied by non-gamers who are open to bias and have knowledge gaps. However, in order to truly understand gaming addiction, it is the game developer who must act as the researcher and embrace ethical wisdom. Thus, it is the game

developer's duty to understand why men are more prone to game addiction than women, why do they keep playing, and what can we as the developer do to ease their burden.

Looking forward, I encourage future researchers, gamers or not, to build upon this paper. There are still many topics within gaming addiction that need addressed, and I hope this paper has provide the building blocks need to expand on it. We know why teens and young adults are becoming hooked into video games, and we know how gambling plays a part in it. Now let's focus on treatment, how can we apply the treatment sued for gambling to addicts to people suffering gaming addiction? What is the best way to apply this kind of treatment, and how can we better encourage gamers to seek help if they feel they need it. In the future, I hope to see more awareness towards gaming addiction, so we may lessen its harm.

References

André, F., Broman, N., Håkansson, A., & Claesdotter-Knutsson, E. (2020). Gaming addiction, problematic gaming and engaged gaming – prevalence and associated characteristics. *Addictive Behaviors Reports, 12*, 100324–100324. https://doi.org/10.1016/j.abrep.2020.100324

Hew, J.-J., Lee, V.-H., T'ng, S.-T., Tan, G. W.-H., Ooi, K.-B., & Dwivedi, Y. K. (2024). Are Online Mobile Gamers Really Happy? On the Suppressor Role of Online Game Addiction. *Information Systems Frontiers, 26*(1), 217–249. https://doi.org/10.1007/s10796-023-10377-7

Inaguma, T., Misumi, S., Funatogawa, T., Nemoto, T., Harima, H., & Mizuno, M. (2024). Does the loot box open the door to addiction? A case report of gaming disorder with high charges for loot box purchases. *PCN Reports*, *3*(1), e167-n/a. https://doi.org/10.1002/pcn5.167

Karhulahti, V.-M. (2024). Vitality structures in "addictive" game design. *Open Research Europe*, *4*, 47-. https://doi.org/10.12688/openreseurope.17177.2

Klemm, C., & Pieters, W. (2017). Game mechanics and technological mediation: an ethical perspective on the effects of MMORPG's. *Ethics and Information Technology, 19*(2), 81–93. https://doi.org/10.1007/s10676-017-9416-6

Yilmaz, V., & Tunca, B. (2024). Structural model proposal to explain online game addiction. *Entertainment Computing, 48*, 100611-. https://doi.org/10.1016/j.entcom.2023.100611